Each group will work on one project and consist of four members. Groups may choose a project other than those listed below, depending on their research interests. Students may also modify or extend the proposed project titles as they see fit.

Area 1: Nature-Inspired Catalysis and Reaction Engineering

1. Enzyme-Inspired Catalytic Reaction Engineering for Low-Temperature Conversion of Starch into Glucose for sustainable food processing
2. Comparative Design of CSTR and PFR Systems for Enzymatic Starch Hydrolysis in Glucose Syrup Production
3. Kinetic Modelling and Reactor Optimization of Enzymatic Biomass Conversion under Mild Operating Conditions
4. Modelling Internal Diffusion and Catalyst Effectiveness in Porous Catalysts for Sustainable Biomass Conversion
5. Kinetic Modelling and Packed-Bed Reactor Design for Catalytic Upgrading of Biomass-Derived Vapours into Valuable Chemicals

Area 2: Biomass Conversion and Waste Valorization

6. Kinetic Modelling and Life-Cycle Assessment of Biochar Production from Rice Husk and Jute Residues
7. Comparative Reactor Design for Catalytic and Non-Catalytic Pyrolysis of Agricultural Residues
8. Process Simulation and Energy Optimization of Catalytic Conversion of Biomass-Derived Compounds
9. Nature-Inspired Biomass Conversion Pathways for Sustainable Chemical Production
10. Comparative Life-Cycle Assessment of Agricultural Residue Conversion Technologies

Area 3: Natural Food Processing and Minimally Processed Foods

11. Kinetic Modelling and Process Optimization of Enzyme-Assisted Food Processing
12. Process Modelling and Life-Cycle Assessment of Plant-Mucilage-Assisted Sugarcane Juice Clarification
13. Energy and Mass Balance Analysis of Low-Energy Sugarcane Juice Concentration and Jaggery Production
14. Kinetic Modelling and Isothermal Reactor Design for Enzymatic Starch Conversion into Glucose
15. Comparative Process and Environmental Assessment of Jaggery, Minimally Refined Sugar and Refined Sugar Production

Area 4: Natural Cosmetics and Bio-Based Personal Care

16. Kinetic Modelling and Optimization of Botanical Compound Extraction for Natural Cosmetics
17. Process Simulation and Life-Cycle Assessment of Low-Temperature Extraction of Plant-Derived Bioactive Compounds
18. Degradation Kinetics and Process Optimization for Aloe Vera Gel Stabilization
19. Reaction Engineering and Resource Optimization in Plant-Oil-Based Soap Production
20. Comparative Life-Cycle Assessment of Conventional and Low-Impact Processing Routes for Natural Cosmetic Ingredients

Area 5: Permaculture, Nutrient Cycling and Environmental Processes

21. Mass Balance Modelling and Life-Cycle Assessment of Organic Waste Conversion for Circular Agriculture
22. Kinetic Modelling and Process Optimization of Organic Waste Stabilization during Composting
23. Process Design and Environmental Assessment of Nutrient Recovery from Agricultural Waste Streams

24. Comparative CSTR and PFR Design for Biological Wastewater Treatment
25. Resource and Energy Balance Modelling of Permaculture-Inspired Agricultural Production Systems

Area 6: Nature-Inspired Reactor Design and Process Intensification (Applications can be tuned based on one's interest.)

26. Nature-Inspired Hierarchical Flow Networks for Improved Reactant Distribution in Catalytic Reactors
27. Kinetic and Mass-Transfer Modelling of Porous Catalysts for Improved Reactor Performance
28. Comparative Isothermal Design of Conventional and Nature-Inspired Packed-Bed Reactors
29. Bio-Inspired Reactor Configurations for Process Intensification and Energy Efficiency
30. Integrating Chemical Reaction Engineering and Life-Cycle Assessment for Sustainable Reactor Design

How to conduct these projects

Every project should use a structured research methodology, even if no experiments are conducted.

Step 1 — Literature data collection

Collect published reaction-rate expressions, kinetic parameters, operating conditions, conversion data, catalyst properties and process energy requirements.

Step 2 — Reaction engineering calculations

Derive mole balances, rate equations and design equations. Calculate conversion, residence time, reactor volume or catalyst mass using Fogler's methodology.

Step 3 — Simulation and optimization

Use Aspen Plus, Python or Excel to compare reactor configurations, vary operating conditions and conduct sensitivity analysis. Simulation can be optional when analytical calculations are sufficient.

Step 4 — Sustainability assessment

Estimate energy consumption, material requirements, waste generation and environmental impacts using transparent assumptions and published life-cycle inventory data.